

Adaptive neural design for permanent magnet synchronous motor with asymmetric constraints and input dead-zone[☆]

Tianhao Fei^a, Yongliang Yang^a,^{*} Chunyu Zheng^b, Hu Zhang^b, Guofeng Yuan^b

^a School of Automation and Electrical Engineering, University of Science and Technology Beijing, Beijing 100083, China

^b Inverter Technologies Engineering Research Center of Beijing, North China University of Technology, Beijing 100144, China

ARTICLE INFO

Communicated by Y. Wang

Keywords:

Permanent magnet synchronous motor
Adaptive Nussbaum design
Barrier Lyapunov function
Unknown control gains
Stochastic disturbance

ABSTRACT

Permanent magnet synchronous motors (PMSMs) with unknown parameters, stochastic disturbance and input nonlinearity are challenging, especially under safety considerations. This paper develops a novel adaptive Nussbaum design scheme for the PMSMs subject to input dead-zones and unknown control gains. The neural network intelligent control technique is combined with the Nussbaum design to compensate for the system uncertainty and unknown control direction. Also, the input dead-zone is taken into consideration within the Nussbaum design framework. To further consider the asymmetric and time-varying constraints, the barrier Lyapunov function is employed to deal with the safety constraints on the electric angular velocity. Theoretical discussions on the stability of the tracking error for PMSMs is investigated using the adaptive Nussbaum design. Simulation studies verify the presented design schemes.

1. Introduction

Permanent Magnet Synchronous Motor (PMSM) offers advantages such as high efficiency, high power density, excellent dynamic performance, low noise, and reliable operation [1]. By using permanent magnets to generate the magnetic field, PMSM eliminates the excitation losses typically found in traditional motors. This not only enables PMSM to deliver higher power output in a compact size, but also ensures reliable speed control and robust performance. These attributes make PMSM widely used in various applications in real life. In robots [2], PMSM provides precise speed control and high torque density, essential for the smooth and accurate movement of robotic joints. In electric vehicles (EVs), its high efficiency and power density contribute to better energy utilization and longer driving ranges, making it an ideal choice for modern EVs [3]. In computer numerically-controlled (CNC) machine tools [4], the precision and reliability of PMSM are crucial for accurate machining processes. Beyond these, PMSM is also employed in wind energy conversion systems [5] with high efficiency and reliability. In automation industry such as electrical submersible pump [6] and flywheel energy system [7], PMSM is widely applied for precise position and speed control with low cost. However, due to the multiple mutually coupled parameters and high nonlinearity, the control of PMSMs is a challenging task [8]. Recent development

in neural networks contributes to intelligent control, which can efficiently deal with complex dynamical systems with uncertainty in engineering systems, such as robotic manipulators [9], and underwater vehicles [10], to name a few. Meanwhile, stochastic behaviors, such as voltage surges, current oscillation, torque fluctuation and other disturbances usually occur in PMSMs [11,12]. These chaotic disturbance can significantly influence the stability of PMSM systems [13] and increase the complexity of design and analysis [14,15]. Therefore, considering stochastic nature of the disturbances in the PMSM dynamics, this paper aims to develop an adaptive neural control scheme for the stochastic PMSMs model with unknown parameters.

In PMSMs, there are multiple parameters involved in the modeling and control design, where the unknown parameters bring a lot challenges. The control gain has a great impact on the control design procedures. For example, in PMSM with unknown stator inductance, the voltage input is difficult to be directly determined. Nussbaum method, an efficient adaptive design scheme, can be employed to compensate for the unknown and time-varying control gain [16,17], which has been widely applied to robotics [18], multi-agent systems [19], to name a few. Chen et al. proposed a novel semi-Nussbaum-based adaptive fuzzy control design to achieve asymptotic stability in the probability sense [20]. Ma et al. combined the adaptive Nussbaum design with the Fourier series expansion to address the nonlinear systems

[☆] This work was supported by the National Natural Science Foundation of China under Grants No. 62373039.

^{*} Corresponding author.

E-mail addresses: m202420790@xs.ustb.edu.cn (T. Fei), yangyongliang@ieee.org (Y. Yang), zhengchunyu@ncut.edu.cn (C. Zheng), zh@ncut.edu.cn (H. Zhang), ygf@ncut.edu.cn (G. Yuan).

<https://doi.org/10.1016/j.neucom.2025.130294>

Received 1 January 2025; Received in revised form 20 February 2025; Accepted 16 April 2025

Available online 27 April 2025

0925-2312/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

with periodic disturbances [21]. However, in the above-mentioned adaptive Nussbaum designs, the input nonlinearity is not taken into account. Typical input nonlinearity, such as saturation [22] and backlash [23], can be commonly found in engineering systems. In this paper, we investigate the issue of unknown control gain and input dead-zone simultaneously.

In engineering applications, safety is a critical performance to be considered to avoid catastrophe and reduce financial risk. The key states in a complex system are usually confined to a certain range. For example, for a given robotic manipulator, the joint angle is required to be within a certain bound to avoid undesired behavior. Typical constraints can be classified as symmetric and asymmetric, which can be efficiently tackled by barrier Lyapunov function (BLF) design. Ren et al. introduced the symmetric BLF design to ensure that the arguments of the unknown functions remain within a compact set [24]. Fang et al. proposed the asymmetric BLF with a power integrator technique to ensure that the system state converges to the origin in finite time and the constraint on system output is not violated [25]. However, these symmetric and asymmetric constraints are time-invariant constraints. The time-varying constraints for PMSM are critical for engineering applications. In this paper, a more general asymmetric and time-varying type of constraint on the electric angular velocity is discussed in detail. To summarize, the contribution of this paper can be collected as follows.

1. The Nussbaum design in [26–28] for systems with a single unknown control direction, which cannot be directly applied to PMSM due to existence of multiple unknown control directions. In contrast, a novel Nussbaum-type function is developed to decouple the multiple unknown control directions in stochastic PMSM dynamics. Accordingly, the novel Nussbaum-type function is combined with the BLF to diminish the effect of unknown control directions on the output constraints.
2. In existing Nussbaum designs [29,30], the constraints on system input and output are considered separately. In PMSM, the electric angular velocity is desired to be within certain limits for safe operation, while the control inputs are subject to dead-zone mapping due to the voltage drop in the PMSM drive circuit component. Therefore, this paper takes the input dead-zone and output constraints into account for PMSM.

The remainder of this paper is organized as follows. Sections 2 and 3 present the preliminaries and problem formulation, respectively. The adaptive backstepping control design is investigated in Section 4 in detail and the stability of the proposed scheme is analyzed in Section 5. In Section 6, the simulation studies of the presented adaptive Nussbaum design for PMSM is implemented. Finally, the concluding remarks are made in Section 7.

2. Preliminary knowledge

2.1. Stochastic systems

Consider the continuous-time stochastic dynamical system as

$$dx = [f(x) + g(x)u]dt + h(x)d\omega, \quad (1)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the control input ω denotes an r -dimensional standard Wiener process defined on a complete probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, P)$ with sample space Ω , σ -field $\mathcal{F}$, filtration $\{\mathcal{F}_t\}_{t \geq 0}$, and probability measure P . $f(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $g(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ and $h(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times r}$ are locally Lipschitz. In this paper, the function $g(\cdot)$ is called the control direction, which is assumed to be unknown and satisfies $0 < G_{\min} \leq |g| \leq G_{\max}$. The following definitions and lemmas are required for the subsequent discussions.

Definition 1 ([31]). For any given second-order continuously differentiable function $V(x) : \mathbb{R}^n \rightarrow \mathbb{R}_+$, associated with the stochastic differential Eq. (1), define the differential operator $\mathcal{L}$ as follows:

$$\mathcal{L}V = \frac{\partial V}{\partial x} F(x) + \frac{1}{2} \text{Tr} \left(h \frac{\partial^2 V}{\partial x^2} h^\top \right), \quad (2)$$

where $F(x) = f(x) + g(x)u$ and $\text{Tr}(\cdot)$ is the matrix trace.

Lemma 1 ([32]). For any positive scalar $\mathcal{Y}$, if any $z \in \mathbb{R}$ satisfies $|z| < \mathcal{Y}$, then, the following inequality holds

$$\log \frac{\mathcal{Y}^{2p}}{\mathcal{Y}^{2p} - z^{2p}} \leq \frac{z^{2p}}{\mathcal{Y}^{2p} - z^{2p}}, \quad (3)$$

where $\log(\cdot)$ represents the natural logarithm and p is a positive integer.

Lemma 2 ([33]). For any positive real numbers ς , ϑ , v and any variables $\mathcal{A}$, $\mathcal{B}$, the following inequality exists

$$|\mathcal{A}|^\varsigma |\mathcal{B}|^\vartheta \leq \frac{\varsigma}{\varsigma + \vartheta} v |\mathcal{A}|^{\varsigma+\vartheta} + \frac{\vartheta}{\varsigma + \vartheta} v^{-\frac{\varsigma}{\vartheta}} |\mathcal{B}|^{\varsigma+\vartheta}. \quad (4)$$

For tracking problem of system (1), the control objective is to track a desired trajectory as the time evolves, while guaranteeing that all the closed-loop signals are bounded almost surely and the tracking error never violates the constraints. To keep the tracking error satisfying the given constraints, the BLF is employed as defined in the following.

Definition 2 ([34]). The function $V(\cdot) : \mathcal{D} \rightarrow \mathbb{R}$ defined on an open set $\mathcal{D} \subset \mathbb{R}^n$ containing the origin with boundary $\partial\mathcal{D}$ is called a BLF for the stochastic system (1) if

- $V(x)$ is continuous, positive definite and has continuous second-order partial derivatives for $\forall x \in \mathcal{D}$,
- $V(x) \rightarrow \infty$ as $x \rightarrow \partial\mathcal{D}$, and
- there exists a positive constant b such that $\mathbb{E}[V(x)] \leq b$, $\forall t \geq 0$.

The BLF is efficient to deal with both time-varying and time-invariant constraints on dynamical systems. Typical BLF designs for deterministic dynamical system are tan-type [35] and integral-type BLF [36]. In this paper, we consider the BLF design $V(x) = \frac{1}{4} \log \left(\frac{1}{1-x^4} \right)$ for stochastic dynamical system.

2.2. Nussbaum design

The Nussbaum design is widely used when the control direction in a dynamical system is unknown. In the Nussbaum design, the key concept is the Nussbaum-type function defined as follows.

Definition 3 ([16]). The Nussbaum-type function $N(\cdot)$ is defined as an even smooth function with the properties

$$\begin{aligned} \limsup_{s \rightarrow \infty} \frac{1}{s} \int_0^s N(\xi) d\xi &= \infty, \\ \liminf_{s \rightarrow \infty} \frac{1}{s} \int_0^s N(\xi) d\xi &= -\infty. \end{aligned} \quad (5)$$

The commonly adopted Nussbaum-type functions are selected as $N(\xi) = \xi^2 \sin(\xi)$, $N(\xi) = \xi^2 \cos(\xi)$ and $N(\xi) = e^{\xi^2} \cos(\frac{\pi}{2}\xi)$. With these above Nussbaum-type functions, the corresponding stability theorem can be found as follows.

Lemma 3 ([37]). Consider the stochastic system (1). Suppose that there exist a smooth signal $\xi(t)$ defined on $[0, t_f]$ and a positive definite function $V(x) \in C^2$ satisfying $k_1(\|x\|) \leq V(x) \leq k_2(\|x\|)$ with $k_1(\cdot), k_1(\cdot) \in \mathcal{K}_\infty$. If $V(x)$ and ξ satisfy

$$\mathcal{L}V(x(t)) \leq -C_1 V(x(t)) + C_2 + [1 + gN(\xi)]\dot{\xi}, \quad (6)$$

with the Nussbaum-type function $N(\cdot)$ and positive constants C_1, C_2 . Then, $\mathbb{E}[V(x)]$ is bounded on $[0, t_f]$.

As discussed in [16], Lemma 3 cannot be directly applied to systems with multiple unknown control directions. To deal with this issue, this paper considers the Nussbaum-type function

$$N_j(\xi) = 2\xi e^{\xi^2} \cos(2^{j-1}\xi) - 2^{j-1} (e^{\xi^2} - 1) \sin(2^{j-1}\xi), \quad (7)$$

for stochastic dynamical system

$$\begin{aligned} dx_j &= (f_j(\bar{x}_j) + g_j(\bar{x}_j)u_{j+1}) dt + h_j(\bar{x}_j)d\omega_j, j = 1, 2, \dots, n-1, \\ dx_n &= (f_n(\bar{x}_n) + g_n(\bar{x}_n)u) dt + h_n(\bar{x}_n)d\omega_n, \end{aligned} \quad (8)$$

where $\bar{x}_j = [x_1, x_2, \dots, x_j]^\top \in \mathbb{R}^j$ is the system state, $x_j \in \mathbb{R}$ is the j -th subsystem state, $u \in \mathbb{R}$ is the control input, ω_j is an one-dimensional standard Wiener process. $f_j : \mathbb{R}^j \rightarrow \mathbb{R}$, $g_j : \mathbb{R}^j \rightarrow \mathbb{R}$ and $h_j : \mathbb{R}^j \rightarrow \mathbb{R}$ are locally Lipschitz, and $0 < G_{\min,j} \leq |g_j| \leq G_{\max,j}$. With the Nussbaum-type function considered in (7), the multiple unknown control directions can be directly tackled by the Nussbaum design, as discussed follows.

Lemma 4. Consider the stochastic dynamical system (8) with multiple bounded control directions g_j for $j = 1, 2, \dots, n$. Suppose that there exists a group of smooth signals $\xi_j(t)$ and a positive definite function $V(x) \in C^2$ satisfying $\chi_1(\|x\|) \leq V(x) \leq \chi_2(\|x\|)$ with $\chi_1(\cdot), \chi_2(\cdot) \in \mathcal{K}_\infty$. If $V(x)$ and $\xi_j(t)$ satisfy

$$\mathcal{L}V(x) \leq -\varpi_1 V(x) + \varpi_2 + \sum_{j=1}^n [g_j N_j(\xi_j) + 1] \dot{\xi}_j. \quad (9)$$

where $N_j(\cdot)$ is a Nussbaum-type function for $j = 1, 2, \dots, N$. Then, one can obtain

$$\mathbb{E}[V(x)] \leq V(x(0))e^{-\varpi_1 t} + \mathcal{E} + \frac{\varpi_2}{\varpi_1}, \quad (10)$$

with positive constants ϖ_1 , ϖ_2 , and $\mathcal{E}$.

Proof. According to [38], the conditions on $f(\cdot)$ and $g(\cdot)$ ensure the existence and uniqueness of a maximal solution $x(t)$ on the time interval $[0, t_f)$, which implies that $V(x(t))$ exists for all $t \in [0, t_f)$. Define $\mathcal{V}(x, t) = V(x)e^{\varpi_1 t}$, one has

$$\begin{aligned} \mathcal{L}\mathcal{V}(x, t) &= \mathcal{L}[V(x)e^{\varpi_1 t}] \\ &= e^{\varpi_1 t} \mathcal{L}V(x) + \varpi_1 V(x)e^{\varpi_1 t}. \end{aligned} \quad (11)$$

Based on the inequality in (9), one has

$$\mathcal{L}\mathcal{V}(x, t) \leq \varpi_2 e^{\varpi_1 t} + e^{\varpi_1 t} \sum_{j=1}^n [g_j N_j(\xi_j) + 1] \dot{\xi}_j. \quad (12)$$

Take expectations on the time interval $[0, t]$, one has

$$\begin{aligned} \mathbb{E}[\mathcal{V}(x, t)] &= \mathbb{E}\left[\int_0^t \mathcal{L}\mathcal{V}(x, \tau) d\tau\right] \\ &\leq \mathbb{E}\left\{\int_0^t e^{\varpi_1 \tau} \left[\varpi_2 + \sum_{j=1}^n [g_j N_j(\xi_j) + 1] \dot{\xi}_j\right] d\tau\right\}. \end{aligned}$$

Denote $\varpi(t) = \int_0^t \varpi_2 e^{\varpi_1 \tau} d\tau = \frac{\varpi_2}{\varpi_1} (e^{\varpi_1 t} - 1)$. Then, the above fact implies that

$$\begin{aligned} \mathbb{E}[\mathcal{V}(x, t)] &\leq \int_0^t e^{\varpi_1 \tau} \sum_{j=1}^n [g_j N_j(\xi_j) + 1] \dot{\xi}_j d\tau \\ &\quad + \varpi(t) + \mathbb{E}[\mathcal{V}(x(0), 0)]. \end{aligned} \quad (13)$$

Based on the definition of $\mathcal{V}(x, t)$, one can obtain the following fact about $V(x)$

$$\begin{aligned} \mathbb{E}[V(x)] &\leq \int_0^t e^{-\varpi_1(t-\tau)} \sum_{j=1}^n [g_j N_j(\xi_j) + 1] \dot{\xi}_j d\tau \\ &\quad + \varpi(t) e^{-\varpi_1 t} + \mathbb{E}[V(x(0))] e^{-\varpi_1 t}. \end{aligned} \quad (14)$$

In the next, the boundedness of ξ_j and integral term in (14) is proved by contradiction. Assuming ξ_j is unbounded, based on [39], one can find

that $\mathbb{E}[V(x)] \leq -\infty$, which contradicts with the positive definiteness of $V(x)$. Thus, ξ_j is bounded. Accordingly, $\int_0^t e^{-\varpi_1(t-\tau)} \sum_{j=1}^n [g_j N_j(\xi_j) + 1] \dot{\xi}_j d\tau$ is also bounded such that

$$\left| \int_0^t e^{-\varpi_1(t-\tau)} \sum_{j=1}^n [g_j N_j(\xi_j) + 1] \dot{\xi}_j d\tau \right| \leq \mathcal{E}, \quad (15)$$

with positive constant $\mathcal{E}$. Also, the term $\varpi(t) e^{-\varpi_1 t}$ is bounded, i.e., $0 \leq \varpi(t) e^{-\varpi_1 t} < \frac{\varpi_2}{\varpi_1}$. Therefore, one has

$$\mathbb{E}[V(x)] \leq V(x(0))e^{-\varpi_1 t} + \mathcal{E} + \frac{\varpi_2}{\varpi_1}. \quad (16)$$

This completes the proof. $\square$

2.3. Radials basis function neural networks

For unknown functions in dynamical system, function approximation techniques can be employed, such as the radial basis function neural networks (RBFNNs). For example, to approximate the unknown function $F(\cdot) : \mathcal{U} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, the RBFNN is considered as

$$\hat{F}(x) = W^\top \varphi(x), \quad (17)$$

with the weight vector $W \in \mathbb{R}^N$ and the radial basis function $\varphi(x) : \mathcal{U} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^N$ defined as

$$\begin{aligned} \varphi(x) &= [\varphi_1(x), \dots, \varphi_N(x)]^\top, \\ \varphi_i(x) &= \exp\left[-\frac{(x - \mu_i)^\top (x - \mu_i)}{2\vartheta_i^2}\right], \end{aligned} \quad (18)$$

where $\mu_i = [\mu_{i1}, \mu_{i2}, \dots, \mu_{in}]^\top \in \mathcal{U}$ is the center and ϑ_i is the width of the Gaussian function. The function approximation accuracy of the RBFNN is investigated in the following lemma.

Lemma 5 ([40]). Consider a continuous function $F(\cdot) : \mathcal{U} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. Given arbitrary accuracy $\epsilon > 0$, there exists an RBFNN in the form of (17) with sufficient hidden nodes number N such that the function approximation residual $\delta(x)$ is bounded on $\mathcal{U}$ in the sense that $\sup_{x \in \mathcal{U}} |\delta(x)| \leq \epsilon$, where

$$\begin{aligned} \delta(x) &= F(x) - W^{*\top} \varphi(x), \\ W^* &= \arg \min_{W \in \mathbb{R}^N} \left\{ \sup_{x \in \mathcal{U}} |F(x) - W^\top \varphi(x)| \right\}. \end{aligned} \quad (19)$$

3. Stochastic permanent magnet synchronous motor

The stochastic PMSM model in the d - q axes can be formulated as

$$\begin{aligned} \frac{3}{2} P_n i_q [i_d (L_d - L_q) + \psi_f] &= T_e, \\ (T_e - T_L) dt + d\omega_1 &= \frac{B\omega_e}{P_n} dt + \frac{J}{P_n} d\omega_e, \\ u_q dt + d\omega_2 &= R i_q dt + L_q d i_q + \omega_e (L_d i_d + \psi_f) dt, \\ u_d dt + d\omega_3 &= R i_d dt + L_d d i_d - \omega_e L_q i_q dt. \end{aligned} \quad (20)$$

where i_d, i_q are the currents, u_d, u_q are the voltages, L_d, L_q are the stator inductances in the d - q axes, respectively. B is the friction coefficient, J is the moment of inertia, ω_e is the electric angular velocity, ψ_f is the permanent magnet flux chain, R is the electronic resistance, P_n is the polar logarithm. T_e is the electromagnetic torque and the load torque T_L is bounded as $|T_L| \leq l$ with positive constant l . ω_1, ω_2 and ω_3 are the standard Wiener process in the torque, q -axis voltage and d -axis voltage, respectively. In this paper, the control signals u_d and u_q are subject to the dead-zone, i.e.,

$$u_k(t) = D_k[v_k(t)] = \begin{cases} D_{k,l}[v_k(t)], & v_k(t) \leq b_{k,l} \\ 0, & b_{k,l} < v_k(t) < b_{k,r} \\ D_{k,r}[v_k(t)], & v_k(t) \geq b_{k,r} \end{cases}, \quad (21)$$

where $k = d, q$ is the d - q axis index, $v_k(t)$ is the real control input needs to be determined. $D_{k,l}[v_k]$, $D_{k,r}[v_k]$ are unknown nonlinear mapping and $b_{k,l} < 0$, $b_{k,r} > 0$ are unknown constant parameters.

By rearranging the terms in (20), one can obtain

$$\begin{aligned} d\omega_e &= \left[\frac{B\omega_e}{J} + \frac{3P_n^2(L_d - L_q)}{2J} i_q i_d + \frac{3P_n^2\psi_f}{2J} i_q - \frac{P_n}{J} T_L \right] dt + \frac{P_n}{J} d\omega_1, \\ di_q &= \left(-\frac{R}{L_q} i_q - \frac{L_d}{L_q} \omega_e i_d - \frac{\psi_f}{L_q} \omega_e + \frac{1}{L_q} u_q \right) dt + \frac{1}{L_q} d\omega_2, \\ di_d &= \left(\frac{L_q}{L_d} \omega_e i_q - \frac{R}{L_d} i_d + \frac{1}{L_d} u_d \right) dt + \frac{1}{L_d} d\omega_3. \end{aligned}$$

For state space representation, denote the state x_i and parameters a_i , b_i , c_i , h_i , respectively, as

$$\begin{cases} x_1 = \omega_e, x_2 = i_q, x_3 = i_d \\ a_1 = -\frac{B}{J}, a_2 = \frac{3P_n^2(L_d - L_q)}{2J}, a_3 = -\frac{P_n}{J}, a_4 = \frac{3P_n^2\psi_f}{2J} \\ b_1 = -\frac{R}{L_q}, b_2 = -\frac{L_d}{L_q}, b_3 = -\frac{\psi_f}{L_q}, b_4 = \frac{1}{L_q} \\ c_1 = -\frac{R}{L_d}, c_2 = \frac{L_q}{L_d}, c_3 = \frac{1}{L_d} \\ h_1 = \frac{P_n}{J}, h_2 = \frac{1}{L_q}, h_3 = \frac{1}{L_d} \end{cases}. \quad (22)$$

According to [41], the PMSM control signals u_q and u_d can be equivalently represented as

$$u_d(t) = d_1(t)v_d(t) + \rho_1(t), u_q(t) = d_2(t)v_q(t) + \rho_2(t), \quad (23)$$

with the control input signals v_q and v_d to be designed. Furthermore, $\rho_1(t)$ and $\rho_2(t)$ are the unknown dead-zone dynamics being bounded as $|\rho_i| \leq \rho_{iM}$ and $|d_i| \leq d_{iM}$ with positive constants ρ_{iM} and d_{iM} , $i = 1, 2$. Based on the result in (23), the PMSM system can be expressed in a compact form as

$$\begin{aligned} dx_1 &= (a_1 x_1 + a_2 x_2 x_3 + a_3 T_L + g_1 x_2) dt + h_1 d\omega_1, \\ dx_2 &= [b_1 x_2 + b_2 x_1 x_3 + b_3 x_1 + g_2 (d_1 v_q + \rho_1)] dt \\ &\quad + h_2 d\omega_2, \\ dx_3 &= [c_1 x_3 + c_2 x_1 x_2 + g_3 (d_2 v_d + \rho_2)] dt \\ &\quad + h_3 d\omega_3. \end{aligned} \quad (24)$$

where $g_1 = a_4$, $g_2 = b_4$ and $g_3 = c_3$. Due to the unknown PMSM system parameters, the coefficients g_1 , g_2 and g_3 of the control signals are also unknown, which are regarded as the unknown control gains. The following assumptions are required for the subsequent discussions.

Assumption 1. For each subsystem, the parameter g_j is unknown and satisfies

$$0 < g_{jm} \leq \|g_j\| \leq g_{JM} \quad (25)$$

with positive constants g_{jm} and g_{JM} .

Assumption 2. For the reference signal x_d , there exist positive constants A_0 , A_1 , and A_2 such that x_d , $\dot{x}_d$, and $\ddot{x}_d$ satisfy $|x_d| \leq A_0$, $|\dot{x}_d| \leq A_1$, $|\ddot{x}_d| \leq A_2$.

Remark 1. Assumption 1 is reasonable since g_j never takes the value 0 at any time and being bounded is the necessary condition for the system (24) to be controllable [42,43]. Assumption 2 is also reasonable and common in engineering applications.

Problem 1. Given a trajectory $x_d(t)$, this paper aims to develop the inputs $v_d(t)$ and $v_q(t)$ for system (24) such that

1. the system output x_1 is driven to follow the reference signal $x_d(t)$ to a bounded compact set, and
 2. the tracking error $e_1 = x_1 - x_d$ approaches the origin and satisfies the constraints
- $$-k_a(t) < e_1(t) < k_b(t) \quad (26)$$

for all $t \geq 0$, where $k_a(t)$ and $k_b(t)$ represent the asymmetric time-varying output constraints satisfying $0 < k_a(t) \leq \kappa_a$ and $0 < k_b(t) \leq \kappa_b$ with positive constants κ_a and κ_b .

4. Adaptive backstepping control design

In the section, an adaptive Nussbaum control design is proposed with backstepping scheme as shown in Fig. 1, where the virtual control output s_1 acts on the first subsystem by the dynamic surface control (DSC) filter output α_1 . For the second and the third subsystem, the control signals v_q and v_d are applied to the PMSM system (24) through the dead-zone outputs u_q and u_d .

The control input is investigated based on the state coordinate transformation with

$$\begin{aligned} e_1 &= x_1 - x_d, e_2 = x_2 - \alpha_1, e_3 = x_3, z_1 = \alpha_1 - s_1, \\ \tilde{a}_2 &= a_2 - \hat{a}_2, \tilde{b}_2 = b_2 - \hat{b}_2, \tilde{\theta}_i = \theta_i - \hat{\theta}_i, \end{aligned} \quad (27)$$

where $i = 1, 2, 3$, x_d is the reference signal satisfying assumption 2, α_1 is the DSC filter internal state, s_1 is the virtual control input design, z_1 is the DSC filter error, $\hat{a}_2$, $\hat{b}_2$ and $\hat{\theta}_i$ are the adaptive compensation for the unknown parameters a_2 , b_2 and θ_i , respectively. Accordingly, $\tilde{a}_2$, $\tilde{b}_2$ and $\tilde{\theta}_i$ are the compensation errors.

4.1. Time-varying constraints transformation

For the asymmetric constraints (26), we consider the following the error transformation

$$\begin{aligned} \eta_1 &= q\eta_b + (1-q)\eta_a, \\ \eta_a &= \frac{e_1}{k_a}, \eta_b = \frac{e_1}{k_b}, \\ q &= \begin{cases} 1, & e_1 > 0 \\ 0, & \text{otherwise} \end{cases}. \end{aligned} \quad (28)$$

From above error transformation, one can observe that,

- $\eta_1 \rightarrow -1$ as $e_1 \rightarrow -k_a$.
- $\eta_1 \rightarrow 1$ as $e_1 \rightarrow k_b$.

Therefore, $\eta_1 \in (-1, 1)$ implies that the constraints (26) are not violated and $\eta_1 \rightarrow 0$ as $e_1 \rightarrow 0$. The error transformation (28) contributes to simplify the time-varying asymmetric constraints as time-invariant symmetric constraints.

Remark 2. To prevent the tracking error from violating the time-varying constraints, the BLF in the form of $V_c(\eta_1) = \frac{1}{4} \log \frac{1}{1-\eta_1^4}$ is employed. By ensuring the boundedness of $V_c(\eta_1)$, one can guarantee that $\eta_1 \in (-1, 1)$, which further implies that the constraints of the tracking error e_1 are never violated.

4.2. Step 1

Differentiating e_1 yields

$$de_1 = f_{e_1} dt + h_1 d\omega_1, \quad (29)$$

where

$$f_{e_1} = a_1 x_1 + a_2 x_2 x_3 + a_3 T_L + g_1 e_2 + g_1 s_1 + g_1 z_1 - \dot{x}_d.$$

The adaptive virtual control s_1 is designed as

$$s_1 = \frac{N_1(\beta_1)\psi_1}{r_1 r_3}, \quad (30)$$

where β_1 is the adaptive Nussbaum gain designed as

$$\begin{aligned} \dot{\beta}_1 &= \xi_1 \psi_1, \\ \xi_1 &= \frac{q\eta_b^3}{k_b(1-\eta_b^4)} + \frac{(1-q)\eta_a^3}{k_a(1-\eta_a^4)}, \end{aligned}$$

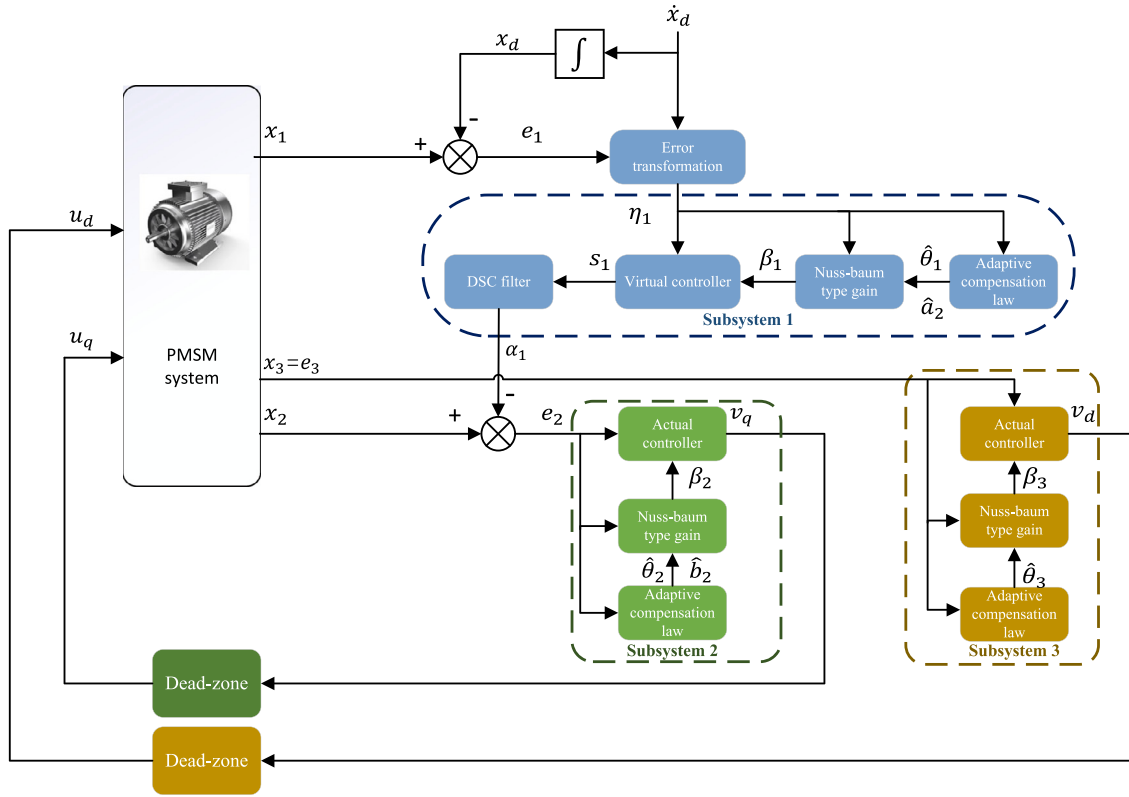


Fig. 1. The flowchart of the proposed control scheme.

$$\begin{aligned}\psi_1 &= k_1 e_1 + \frac{\xi_1^2 r_3^2 \hat{\theta}_1 \varphi_1^\top \varphi_1}{2} + r_1 r_3 \hat{a}_2 x_2 x_3, \\ &\quad - r_1 r_3 \dot{x}_d - r_2 r_3 e_1 + 2 \xi_1^2 r_3^2 \\ r_1 &= \frac{q}{k_b} + \frac{1-q}{k_a}, \\ r_2 &= \frac{q k_b}{k_b^2} + \frac{(1-q) \dot{k}_a}{k_a^2}, \\ r_3 &= q k_b + (1-q) k_a,\end{aligned}\quad (31)$$

with the following adaptive compensation

$$\begin{aligned}\dot{\hat{\theta}}_1 &= \frac{\xi_1^2 r_3^2 \varphi_1^\top \varphi_1}{2} - \gamma_{\theta_1} \hat{\theta}_1, \\ \dot{\hat{a}}_2 &= \xi_1 r_1 r_3 x_2 x_3 - \gamma_{a_2} \hat{a}_2.\end{aligned}\quad (32)$$

The DSC filter is designed as

$$\tau_1 \dot{\alpha}_1 = s_1 - \alpha_1.$$

From above notations, the boundedness of k_a and k_b implies that r_1 is also bounded, i.e., there exists a positive constant $\bar{r}_1$ such that $0 < r_1 \leq \bar{r}_1$.

Based on the design of s_1 , one can obtain the derivative of s_1 as

$$\begin{aligned}\dot{s}_1 &= \frac{r_1 r_3 [\dot{\beta}_1 \dot{N}_1(\beta_1) \psi_1 + N_1(\beta_1) \dot{\psi}_1]}{(r_1 r_3)^2} \\ &\quad - \frac{(\dot{r}_1 r_3 + r_1 \dot{r}_3) N_1(\beta_1) \psi_1}{(r_1 r_3)^2}.\end{aligned}\quad (33)$$

Define the sets

$$\begin{aligned}\Omega_1 &= \left\{ \mathcal{X}_1 \left| \frac{1}{4} \ln \frac{1}{1-\eta_1^4} + \frac{1}{4} z_1^4 + \frac{1}{2} \bar{\theta}_1^2 + \frac{1}{2} \bar{a}_2^2 \leq R_1 \right. \right\}, \\ \Omega_{x_d} &= \left\{ (x_d, \dot{x}_d, \ddot{x}_d) \left| \frac{1}{2} x_d^2 + \frac{1}{2} \dot{x}_d^2 + \frac{1}{2} \ddot{x}_d^2 \leq R_0 \right. \right\}.\end{aligned}\quad (34)$$

where $\mathcal{X}_1 = (e_1, z_1, \bar{\theta}_1, \bar{a}_2)$, $R_0 > 0$, and $R_1 > 0$. On the compact set $\Omega_1 \times \Omega_{x_d}$, it can be noted that s_1 is continuous and $|\dot{s}_1| \leq M_1$.

4.3. Step 2

Differentiating e_2 yields

$$d e_2 = f_{e_2} dt + h_2 d \omega_2, \quad (35)$$

where $f_{e_2} = b_1 x_2 + b_2 x_1 x_3 + b_3 x_1 + g_2 d_1 v_q + g_2 \rho_1 - \dot{\alpha}_1$. The adaptive control signal v_q can be designed as

$$v_q = N_2(\beta_2) \psi_2, \quad (36)$$

where β_2 is the adaptive Nussbaum gain designed as

$$\begin{aligned}\beta_2 &= e_2^3 \psi_2, \\ \psi_2 &= \frac{1}{2} e_2^3 \hat{\theta}_2 \varphi_2^\top \varphi_2 + \left(k_2 + \frac{3}{2} \right) e_2 - \dot{\alpha}_1 \\ &\quad + \hat{b}_2 x_1 x_3,\end{aligned}\quad (37)$$

with the following adaptive compensation

$$\begin{aligned}\dot{\hat{\theta}}_2 &= \frac{1}{2} e_2^6 \varphi_2^\top \varphi_2 - \gamma_{\theta_2} \hat{\theta}_2, \\ \dot{\hat{b}}_2 &= e_2^3 x_1 x_3 - \gamma_{b_2} \hat{b}_2.\end{aligned}\quad (38)$$

4.4. Step 3

Differentiating e_3 yields

$$d e_3 = f_{e_3} dt + h_3 d \omega_3, \quad (39)$$

where $f_{e_3} = c_1 x_3 + c_2 x_1 x_2 + g_3 (d_2 v_d + \rho_2)$. The adaptive actual control signal v_d can be designed as

$$v_d = N_3(\beta_3) \psi_3, \quad (40)$$

where β_3 is the adaptive Nussbaum gain designed as

$$\begin{aligned}\dot{\beta}_3 &= e_3^3 \psi_3, \\ \psi_3 &= \frac{1}{2} e_3^3 \hat{\theta}_3 \varphi_3^\top \varphi_3 + \left(k_3 + \frac{3}{2}\right) e_3,\end{aligned}\quad (41)$$

with the following adaptive compensation

$$\dot{\hat{\theta}}_3 = \frac{1}{2} e_3^6 \varphi_3^\top \varphi_3 - \gamma_{\theta_3} \hat{\theta}_3. \quad (42)$$

5. Stability analysis

5.1. Step 1

The derivative of η_1 can be expressed as

$$\begin{aligned}d\eta_1 &= q d\eta_b + (1-q) d\eta_a \\ &= q \left(\frac{de_1}{k_b} - e_1 \frac{dk_b}{k_b^2} \right) + (1-q) \left(\frac{de_1}{k_a} - e_1 \frac{dk_a}{k_a^2} \right) \\ &= q \left(\frac{f_{e_1} dt + h_1 d\omega_1}{k_b} - e_1 \frac{k_b dt}{k_b^2} \right) \\ &\quad + (1-q) \left(\frac{f_{e_1} dt + h_1 d\omega_1}{k_a} - e_1 \frac{k_a dt}{k_a^2} \right) \\ &= f_{\eta_1} dt + H_1 d\omega_1,\end{aligned}\quad (43)$$

where

$$\begin{aligned}f_{\eta_1} &= \frac{q}{k_b} \left(\frac{f_{e_1}}{dt} - e_1 \frac{k_b}{k_b^2} \right) + \frac{1-q}{k_a} \left(\frac{f_{e_1}}{dt} - e_1 \frac{k_a}{k_a^2} \right), \\ H_1 &= h_1 \left(\frac{q}{k_b} + \frac{1-q}{k_a} \right).\end{aligned}$$

Consider the Lyapunov candidate

$$V_1(\eta_1, z_1, \tilde{\theta}_1, \tilde{a}_2) = V_c(\eta_1) + V_d(z_1, \tilde{\theta}_1, \tilde{a}_2),$$

where $V_d(z_1, \tilde{\theta}_1, \tilde{a}_2) = \frac{1}{4} z_1^4 + \frac{1}{2} \tilde{\theta}_1^2 + \frac{1}{2} \tilde{a}_2^2$, $V_c(\eta_1) = \frac{1}{4} \log \left(\frac{1}{1-\eta_1^4} \right)$ is the BLF satisfying $V_c \rightarrow \infty$ as $e_1 \rightarrow -k_a$ or $e_1 \rightarrow k_b$. Based on the transformed error η_1 definition in (28), one can further obtain

$$V_c(\eta_1) = \frac{1}{4} \left[q \log \left(\frac{k_a^4}{k_a^4 - e_1^4} \right) + (1-q) \log \left(\frac{k_b^4}{k_b^4 - e_1^4} \right) \right].$$

Then, one has

$$\mathcal{L}V_1 = z_1^3 \dot{z}_1 + \tilde{\theta}_1 \dot{\tilde{\theta}}_1 + \tilde{a}_2 \dot{\tilde{a}}_2 + \frac{\eta_1^3}{1-\eta_1^4} f_{\eta_1} + \frac{1}{2} \text{Tr} \left\{ H_1^\top \frac{\partial^2 V_1}{\partial \eta_1^2} H_1 \right\}.$$

For the terms $\frac{\eta_1^3}{1-\eta_1^4}$ and f_{η_1} , one has

$$\begin{aligned}\frac{\eta_1^3}{1-\eta_1^4} &= \xi_1 r_3, \\ f_{\eta_1} &= r_1 a_1 x_1 + r_1 a_2 x_2 x_3 + r_1 g_1 (e_2 + s_1 + z_1) \\ &\quad - r_1 \dot{x}_d - e_1 r_2 + r_1 a_3 T_L.\end{aligned}\quad (44)$$

Inserting (44) into $\mathcal{L}V_1$ yields

$$\begin{aligned}\mathcal{L}V_1 &= \xi_1 r_3 (r_1 a_1 x_1 + r_1 a_2 x_2 x_3 - r_1 \dot{x}_d - r_2 e_1) \\ &\quad + \xi_1 r_1 r_3 g_1 e_2 + \xi_1 r_1 r_3 a_3 T_L + \xi_1 r_1 r_3 g_1 z_1 \\ &\quad + \xi_1 r_1 r_3 g_1 s_1 - \frac{z_1^4}{\tau_1} - z_1^3 \dot{s}_1 - \tilde{\theta}_1 \dot{\tilde{\theta}}_1 - \tilde{a}_2 \dot{\tilde{a}}_2 \\ &\quad + \frac{3\eta_1^2 + \eta_1^6}{2(1-\eta_1^4)^2} H_1^2.\end{aligned}\quad (45)$$

For the terms in (45), applying Lemma 2, one has

$$\xi_1 r_1 r_3 g_1 e_2 \leq \frac{\xi_1^2 r_3^2}{2} + \frac{\bar{r}_1^2 g_{1M}^2 e_2^2}{2}$$

$$\leq \frac{\xi_1^2 r_3^2}{2} + \frac{\bar{r}_1^2 g_{1M}^2}{2} \left(\frac{\sigma_e}{2} + \frac{e_2^4}{2\sigma_e} \right),$$

$$\xi_1 r_1 r_3 a_3 T_L \leq \frac{\xi_1^2 r_3^2}{2} + \frac{\bar{r}_1^2 a_3^2 l^2}{2},$$

$$\xi_1 r_1 r_3 g_1 z_1 \leq \frac{\xi_1^2 r_3^2}{2} + \frac{\bar{r}_1^2 g_{1M}^2}{2} z_1^2$$

$$\leq \frac{\xi_1^2 r_3^2}{2} + \frac{\bar{r}_1^2 g_{1M}^2}{2} \left(\frac{\sigma_z}{2} + \frac{1}{2\sigma_z} z_1^4 \right),$$

$$\frac{\eta_1^2 (3 + \eta_1^4)}{2(1 - \eta_1^4)^2} H_1^2 \leq \xi_1 r_3 \left(\frac{\eta_1 (3 + \eta_1^4)^2 H_1^4}{4\varsigma_1 (1 - \eta_1^4)^3} \right) + \frac{\varsigma_1}{4}, \quad (46)$$

where σ_e , σ_z and ς_1 are positive constants. Inserting (46) into (45), one has

$$\begin{aligned}\mathcal{L}V_1 &\leq \xi_1 r_3 \left[F_1(x_1, r_1, \eta_1) + r_1 a_2 x_2 x_3 - r_1 \dot{x}_d \right. \\ &\quad \left. - r_2 e_1 + \frac{3}{2} \xi_1 r_3 \right] + \frac{\bar{r}_1^3 a_3^2 l^2}{2} - \frac{z_1^4}{\tau_1} - z_1^3 \dot{s}_1 \\ &\quad + \xi_1 r_1 r_3 g_1 s_1 + \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_z} z_1^4 + \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_e} e_2^4 + \frac{\varsigma_1}{4} \\ &\quad - \tilde{\theta}_1 \dot{\tilde{\theta}}_1 - \tilde{a}_2 \dot{\tilde{a}}_2 + \frac{\sigma_z \bar{r}_1^2 g_{1M}^2}{4} + \frac{\sigma_e \bar{r}_1^2 g_{1M}^2}{4},\end{aligned}\quad (47)$$

where

$$F_1(x_1, r_1, \eta_1) = r_1 a_1 x_1 + \frac{\eta_1 (3 + \eta_1^4)^2 H_1^4}{4\varsigma_1 (1 - \eta_1^4)^3} \quad (48)$$

is an unknown function due to the terms a_1 and H_1 . To deal with this issue, the unknown function $F_1(x_1, r_1, \eta_1)$ can be expressed using RBFNN based on Lemma 5 as

$$F_1(x_1, r_1, \eta_1) = W_1^\top \varphi_1(x_1, r_1, \eta_1) + \delta_1(x_1, r_1, \eta_1), \quad (49)$$

where W_1 denotes the unknown ideal weight and the residual satisfies $|\delta_1(x_1, r_1, \eta_1)| \leq \varepsilon_1$. Taking the RBFNN (49) into (47) yields

$$\begin{aligned}\mathcal{L}V_1 &\leq \xi_1 r_3 (W_1^\top \varphi_1 + \delta_1 + r_1 a_2 x_2 x_3 - r_1 \dot{x}_d \\ &\quad - r_2 e_1 + \xi_1 r_3) + \xi_1 r_1 r_3 g_1 s_1 - \frac{z_1^4}{\tau_1} + \frac{\bar{r}_1^3 a_3^2 l^2}{2} \\ &\quad + \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_z} z_1^4 - z_1^3 \dot{s}_1 - \tilde{\theta}_1 \dot{\tilde{\theta}}_1 - \tilde{a}_2 \dot{\tilde{a}}_2 + \frac{\varsigma_1}{4} \\ &\quad + \frac{(\sigma_e + \sigma_z) \bar{r}_1^2 g_{1M}^2}{4} + \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_e} e_2^4.\end{aligned}\quad (50)$$

Based on Lemma 2, one has:

$$\begin{aligned}\xi_1 r_3 W_1^\top \varphi_1 &\leq \frac{\xi_1^2 r_3^2 \theta_1^\top \varphi_1}{2} + \frac{1}{2}, \\ \xi_1 r_3 \delta_1 &\leq \frac{\xi_1^2 r_3^2}{2} + \frac{\varepsilon_1^2}{2},\end{aligned}\quad (51)$$

where $\theta_1 = W_1^\top W_1$. Since the ideal RBFNN weight vector W_1 is unknown, θ_1 is also unknown, which can be compensated by the adaptive design in (32).

Adding and subtracting the term $k_1 e_1 \xi_1$ in (50) and inserting the inequalities (51) into (50), one has

$$\begin{aligned}\mathcal{L}V_1 &\leq -k_1 e_1 \xi_1 + \xi_1 \left(k_1 e_1 + \frac{\xi_1 r_3^2 \theta_1^\top \varphi_1}{2} + r_1 r_3 \hat{a}_2 x_2 x_3 \right. \\ &\quad \left. - r_1 r_3 \dot{x}_d - r_2 r_3 e_1 + 2\xi_1 r_3^2 \right) + \xi_1 r_1 r_3 g_1 s_1 + \frac{1 + \varepsilon_1^2}{2} \\ &\quad - \frac{z_1^4}{\tau_1} + \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_z} z_1^4 - z_1^3 \dot{s}_1 + \frac{\varsigma_1}{4} + \frac{(\sigma_e + \sigma_z) \bar{r}_1^2 g_{1M}^2}{4} \\ &\quad - \tilde{\theta}_1 \left(\dot{\tilde{\theta}}_1 - \frac{\xi_1^2 r_3^2 \varphi_1^\top \varphi_1}{2} \right) - \tilde{a}_2 (\dot{\tilde{a}}_2 - \xi_1 r_1 r_3 x_2 x_3)\end{aligned}$$

$$+ \frac{\bar{r}_1^3 a_3^2 l^2}{2} + \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_e} e_2^4. \quad (52)$$

Recall the fact $|\dot{s}_1| \leq M_1$. The term $-z_1^3 \dot{s}_1$ in (52) satisfies

$$-z_1^3 \dot{s}_1 \leq \frac{3M_1^4}{4\zeta_z} z_1^4 + \frac{\zeta_z}{4},$$

where $\zeta_z > 0$. Inserting the above inequality, the virtual control signal s_1 in (30), and the adaptive compensation (32) into (52) yields

$$\begin{aligned} \mathcal{L}V_1 \leq & -k_1 e_1 \xi_1 - \left(\frac{1}{\tau_1} - \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_z} - \frac{3M_1^4}{4\zeta_z} \right) z_1^4 + \frac{\epsilon_1^2}{2} \\ & + \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_e} e_2^4 + [1 + g_1 N_1(\beta_1)] \dot{\beta}_1 + \gamma_{\theta_1} \tilde{\theta}_1 \hat{\theta}_1 \\ & + \gamma_{a_2} \tilde{a}_2 \hat{a}_2 + \frac{\zeta_1}{4} + \frac{(\sigma_e + \sigma_z) \bar{r}_1^2 g_{1M}^2}{4} \\ & + \frac{1 + \bar{r}_1^3 a_3^2 l^2}{2}. \end{aligned} \quad (53)$$

5.2. Step 2

Consider the Lyapunov function as

$$V_2 = \frac{1}{4} e_2^4 + \frac{1}{2} \tilde{\theta}_2^2 + \frac{1}{2} \tilde{b}_2^2. \quad (54)$$

Based on the dynamics of e_2 , one can obtain

$$\begin{aligned} \mathcal{L}V_2 &= e_2^3 f_{e_2} + \tilde{\theta}_2 \dot{\tilde{\theta}}_2 + \tilde{b}_2 \dot{\tilde{b}}_2 + \frac{1}{2} \text{Tr} \left\{ h_2^\top \frac{\partial^2 V_2}{\partial e_2^2} h_2 \right\} \\ &= e_2^3 (b_1 x_2 + b_2 x_1 x_3 + b_3 x_1 + g_2 d_1 v_q \\ &\quad + g_2 \rho_1 - \dot{\alpha}_1) - \tilde{\theta}_2 \dot{\tilde{\theta}}_2 - \tilde{b}_2 \dot{\tilde{b}}_2 + \frac{3}{2} e_2^2 h_2^2 \\ &= e_2^3 (b_1 x_2 + b_2 x_1 x_3 + b_3 x_1 + g_2 d_1 v_q - \dot{\alpha}_1) \\ &\quad + e_2^3 g_2 \rho_1 - \tilde{\theta}_2 \dot{\tilde{\theta}}_2 - \tilde{b}_2 \dot{\tilde{b}}_2 + \frac{3}{2} e_2^2 h_2^2. \end{aligned} \quad (55)$$

For the terms $e_2^2 h_2^2$ and $e_2^3 g_2 \rho_1$ in (55), one has

$$\begin{aligned} e_2^2 h_2^2 &\leq \frac{e_2^4 h_2^4}{2\zeta_2} + \frac{\zeta_2}{2}, \\ e_2^3 g_2 \rho_1 &\leq \frac{3}{4} e_2^4 + \frac{1}{4} g_{2M}^4 \rho_{1M}^4. \end{aligned} \quad (56)$$

Inserting (56) into (55), one has

$$\begin{aligned} \mathcal{L}V_2 \leq & e_2^3 \left[F_3(x_1, x_2, e_2) + b_2 x_1 x_3 + g_2 d_1 v_q \right. \\ & \left. + \frac{3}{4} e_2 + \frac{3e_2 h_2^4}{4\zeta_2} - \dot{\alpha}_1 \right] - \tilde{\theta}_2 \dot{\tilde{\theta}}_2 + \frac{3\zeta_2}{4} \\ & - \tilde{b}_2 \dot{\tilde{b}}_2 + \frac{1}{4} g_{2M}^4 \rho_{1M}^4, \end{aligned} \quad (57)$$

where

$$F_2(x_1, x_2, e_2) = b_1 x_2 + b_3 x_1 + \frac{3e_2 h_2^4}{4\zeta_2}. \quad (58)$$

Based on Lemma 5, the function $F_2(x_1, x_2, e_2)$ can be expressed using the RBFNN as

$$F_2(x_1, x_2, e_2) = W_2^\top \varphi_2(x_1, x_2, e_2) + \delta_2(x_1, x_2, e_2), \quad (59)$$

where W_2 is the unknown ideal weight and the residual satisfies $|\delta_2(x_1, x_2, e_2)| \leq \epsilon_2$. Taking the RBFNN (59) into (57) yields

$$\begin{aligned} \mathcal{L}V_2 \leq & e_2^3 W_2^\top \varphi_2 + e_2^3 \delta_2 + \frac{3\zeta_2}{4} + e_2^3 (b_2 x_1 x_3 + g_2 d_1 v_q \\ & + \frac{3}{4} e_2 - \dot{\alpha}_1) - \tilde{\theta}_2 \dot{\tilde{\theta}}_2 - \tilde{b}_2 \dot{\tilde{b}}_2 + \frac{1}{4} g_{2M}^4 \rho_{1M}^4. \end{aligned} \quad (60)$$

Based on Lemma 2, one has

$$e_2^3 \delta_2 \leq \frac{3}{4} e_2^4 + \frac{1}{4} \epsilon_2^4,$$

$$e_2^3 W_2^\top \varphi_2 \leq \frac{1}{2} e_2^6 \theta_2 \varphi_2^\top \varphi_2 + \frac{1}{2}, \quad (61)$$

where $\theta_2 = W_2^\top W_2$. Since the ideal weight W_2 is unknown, θ_2 is also unknown, which can be compensated by the adaptive design in (38). Inserting (61) into (60), one can obtain

$$\begin{aligned} \mathcal{L}V_2 \leq & -k_2 e_2^4 + e_2^3 \left(\dot{b}_2 x_1 x_3 + \frac{1}{2} e_2^3 \tilde{\theta}_2 \varphi_2^\top \varphi_2 \right. \\ & \left. + \frac{3}{2} e_2 + k_2 e_2 - \dot{\alpha}_1 \right) + g_2 d_1 e_2^3 v_q \\ & - \tilde{\theta}_2 \left(\dot{\tilde{\theta}}_2 - \frac{1}{2} e_2^6 \varphi_2^\top \varphi_2 \right) - \tilde{b}_2 \left(\dot{\tilde{b}}_2 - e_2^3 x_1 x_3 \right) \\ & + \frac{3\zeta_2}{4} + \frac{1}{4} g_{2M}^4 \rho_{1M}^4 + \frac{1}{4} \epsilon_2^4 + \frac{1}{2}. \end{aligned} \quad (62)$$

Substitute the design of v_q in (36) and adaptive compensation in (38), one can obtain

$$\begin{aligned} \mathcal{L}V_2 \leq & -k_2 e_2^4 + \gamma_{\theta_2} \tilde{\theta}_2 \dot{\tilde{\theta}}_2 + \gamma_{b_2} \tilde{b}_2 \dot{\tilde{b}}_2 + \frac{3\zeta_2}{4} + \frac{1}{4} \epsilon_2^4 + \frac{1}{2} \\ & + [1 + g_2 d_1 N_2(\beta_2)] \dot{\beta}_2 + \frac{1}{4} g_{2M}^4 \rho_{1M}^4. \end{aligned} \quad (63)$$

5.3. Step 3

Consider the Lyapunov function $V_3 = \frac{1}{4} e_3^4 + \frac{1}{2} \tilde{\theta}_3^2$. Then, one can obtain

$$\begin{aligned} \mathcal{L}V_3 &= e_3^3 f_{e_3} + \tilde{\theta}_3 \dot{\tilde{\theta}}_3 + \frac{1}{2} \text{Tr} \left\{ h_3^\top \frac{\partial^2 V_3}{\partial e_3^2} h_3 \right\} \\ &= e_3^3 (c_1 x_3 + c_2 x_1 x_2) + g_3 d_2 e_3^3 v_d \\ &\quad + e_3^3 g_3 \rho_2 - \tilde{\theta}_3 \dot{\tilde{\theta}}_3 + \frac{3}{2} e_3^2 h_3^2. \end{aligned} \quad (64)$$

For the terms $e_3^2 h_3^2$ and $e_3^3 g_3 \rho_2$ in (64), there exists inequalities as

$$\begin{aligned} e_3^2 h_3^2 &\leq \frac{e_3^4 h_3^4}{2\zeta_3} + \frac{\zeta_3}{2}, \\ e_3^3 g_3 \rho_2 &\leq \frac{3}{4} e_3^4 + \frac{1}{4} g_{3M}^4 \rho_{2M}^4, \end{aligned} \quad (65)$$

where $\zeta_3 > 0$. Inserting (65) into (64), (64) can be rewritten as

$$\begin{aligned} \mathcal{L}V_3 \leq & e_3^3 \left[F_3(x_1, x_2, e_3) + \frac{3}{4} e_3 \right] - \tilde{\theta}_3 \dot{\tilde{\theta}}_3 \\ & + g_3 d_2 e_3^3 v_d + \frac{1}{4} g_{3M}^4 \rho_{2M}^4 + \frac{3\zeta_3}{4}, \end{aligned} \quad (66)$$

where

$$F_3(x_1, x_2, e_3) = c_1 e_3 + c_2 x_1 x_2 + \frac{3e_3 h_3^4}{4\zeta_3} \quad (67)$$

is an unknown function due to the unknown parameters c_1 , c_2 and h_3 . To deal with this unknown function, the RBFNN is employed to approach the unknown function as

$$\begin{aligned} F_3(x_1, x_2, e_3) &= W_3^\top \varphi_3(x_1, x_2, e_3) + \delta_3(x_1, x_2, e_3), \end{aligned} \quad (68)$$

where W_3 is the unknown ideal weight and the residual satisfies $|\delta_3(x_1, x_2, e_3)| \leq \epsilon_3$. Taking the RBFNN (68) into (69) yields

$$\begin{aligned} \mathcal{L}V_3 \leq & e_3^3 \left(W_3^\top \varphi_3 + \delta_3 + \frac{3}{4} e_3 \right) + g_3 d_2 e_3^3 v_d \\ & - \tilde{\theta}_3 \dot{\tilde{\theta}}_3 + \frac{1}{4} g_{3M}^4 \rho_{2M}^4 + \frac{3\zeta_3}{4}. \end{aligned} \quad (69)$$

According to Lemma 2, one has

$$\begin{aligned} e_3^3 \delta_3 &\leq \frac{3}{4} e_3^4 + \frac{1}{4} \epsilon_3^4, \\ e_3^3 W_3^\top \varphi_3 &\leq \frac{1}{2} e_3^6 \theta_3 \varphi_3^\top \varphi_3 + \frac{1}{2}, \end{aligned} \quad (70)$$

where $\theta_3 = W_3^\top W_3$. Due to the unknown ideal weight W_3 , θ_3 is unknown as well, which can be compensated by the adaptive design in (42). Inserting (70) into (69), one can obtain

$$\mathcal{L}V_3 \leq -k_3 e_3^4 + e_3^3 \left(\frac{1}{2} e_3^3 \tilde{\theta}_3 \varphi_3^\top \varphi_3 + \frac{3}{2} e_3 + k_3 e_3 \right)$$

$$\begin{aligned}
& + g_3 d_2 e_3^3 v_d - \bar{\theta}_3 \left(\dot{\bar{\theta}}_3 - \frac{1}{2} e_3^6 \varphi_3^\top \varphi_3 \right) \\
& + \frac{1}{4} g_{3M}^4 \rho_{2M}^4 + \frac{3\zeta_3}{4} + \frac{1}{4} \epsilon_3^4 + \frac{1}{2}.
\end{aligned} \quad (71)$$

Based on the design of v_d in (40) and adaptive law in (42), one can obtain

$$\begin{aligned}
\mathcal{L}V_3 & \leq -k_3 e_3^4 + \gamma_{\theta_3} \bar{\theta}_3 \dot{\bar{\theta}}_3 + [1 + g_3 d_2 N_3(\bar{\beta}_3)] \dot{\bar{\beta}}_3 \\
& + \frac{1}{4} g_{3M}^4 \rho_{2M}^4 + \frac{3\zeta_3}{4} + \frac{1}{4} \epsilon_3^4 + \frac{1}{2}.
\end{aligned} \quad (72)$$

To this end, the stability analysis of the stochastic PMSM with the presented design in Section 4 can be summarized as follows.

Theorem 1. Consider stochastic PMSM system (20) subject to input dead-zone (21) under assumptions 1 and 2. Suppose that the output error e_1 at initial instant satisfies the constraints (26), i.e.,

$$e_1(0) \in \Omega_{e_1} = \{e_1 : -k_a(0) < e_1(0) < k_b(0)\}.$$

Then, with s_1 designed in (30), v_q designed in (36), v_d designed in (40), and the adaptive compensation laws designed in Section 4, the following facts can be obtained

1. the output tracking error $e_1(t)$ is driven to a compact set defined as

$$\begin{aligned}
\bar{\Omega}_{e_1} & = \left\{ e_1 \left| -k_a \sqrt{1 - e^{-4V(\mathcal{W}(0))e^{-\lambda_1 t} - \frac{4\lambda_2}{\lambda_1} - 4\bar{\mathcal{E}}}} \right. \right. \\
& \leq \mathbb{E}(e_1) \leq k_b \sqrt{1 - e^{-4V(\mathcal{W}(0))e^{-\lambda_1 t} - \frac{4\lambda_2}{\lambda_1} - 4\bar{\mathcal{E}}}} \left. \right\}, \text{ and}
\end{aligned}$$

2. the output error signal $e_1(t)$ never violates the constraints (26).

Proof. For the stochastic PMSM system (24), consider the Lyapunov candidate function as

$$V(\mathcal{W}) = V_1(\eta_1, z_1, \bar{\theta}_1, \bar{a}_2) + V_2(e_2, \bar{\theta}_2, \bar{b}_2) + V_3(e_3, \bar{\theta}_3),$$

where $\mathcal{W} = [\eta_1, e_2, e_3, \bar{\theta}_1, \bar{\theta}_2, \bar{\theta}_3, \bar{a}_2, \bar{b}_2]^\top$. Then, one has

$$\mathcal{L}V = \mathcal{L}V_1 + \mathcal{L}V_2 + \mathcal{L}V_3. \quad (73)$$

Based on the notation of ξ_1 , one can observe that $\xi_1 = \frac{qe_1^3}{k_b^4(1-\eta_b^4)} + \frac{(1-q)e_1^3}{k_a^4(1-\eta_a^4)}$. Then, by Lemmas 1 and 2, there exist inequalities

$$\begin{aligned}
e_1 \xi_1 & = \frac{qe_1^4}{k_b^4(1-\eta_b^4)} + \frac{(1-q)e_1^4}{k_a^4(1-\eta_a^4)} \\
& = \frac{\eta_1^4}{1-\eta_1^4} \geq \log\left(\frac{1}{1-\eta_1^4}\right), \\
\bar{\theta}_i \dot{\bar{\theta}}_i & \leq -\frac{1}{2} \bar{\theta}_i^2 + \frac{1}{2} \theta_i^2, \\
\bar{a}_2 \dot{\bar{a}}_2 & \leq -\frac{1}{2} \bar{a}_2^2 + \frac{1}{2} a_2^2, \\
\bar{b}_2 \dot{\bar{b}}_2 & \leq -\frac{1}{2} \bar{b}_2^2 + \frac{1}{2} b_2^2.
\end{aligned} \quad (74)$$

Inserting each inequality in (74) into (53), (63), (72), and (73), respectively, one can obtain

$$\begin{aligned}
\mathcal{L}V & \leq -\lambda_{1,1} V_1 - \left(k_2 - \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_e}\right) e_2^4 - \frac{\gamma_{\theta_2}}{2} \bar{\theta}_2^2 - \frac{\gamma_{b_2}}{2} \bar{b}_2^2 \\
& - \lambda_{3,1} V_3 + \sum_{j=1}^3 [1 + G_j N_j(\bar{\beta}_j)] \dot{\bar{\beta}}_j + \sum_{j=1}^3 \lambda_{j,2},
\end{aligned} \quad (75)$$

where

$$\lambda_{1,1} = \min \left\{ 4k_1, \frac{4}{\tau_1} - \frac{\bar{r}_1^2 g_{1M}^2}{\sigma_z} - \frac{3M_1^{\frac{4}{3}}}{\zeta_z}, \gamma_{\theta_1}, \gamma_{a_1} \right\},$$

$$\begin{aligned}
\lambda_{1,2} & = \frac{\zeta_1}{4} + \frac{\zeta_z}{4} + \frac{(\sigma_e + \sigma_z) \bar{r}_1^2 g_{1M}^2}{4} + \frac{\bar{r}_1^3 a_1^2 l^2}{2} + \frac{1 + \epsilon_1^2}{2} \\
& + \frac{\gamma_{\theta_1}}{2} \theta_1^2 + \frac{\gamma_{a_2}}{2} a_2^2, \\
\lambda_{2,2} & = \frac{3\zeta_2}{4} + \frac{1}{2} g_{2M}^4 \rho_{1M}^4 + \frac{1}{4} \epsilon_2^4 + \frac{\gamma_{\theta_2}}{2} \theta_2^2 + \frac{\gamma_{b_2}}{2} b_2^2 + \frac{1}{2}, \\
\lambda_{3,1} & = \min \{4k_3, \gamma_{\theta_3}\}, \\
\lambda_{3,2} & = \frac{1}{4} g_{3M}^4 \rho_{2M}^4 + \frac{3\zeta_3}{4} + \frac{1}{4} \epsilon_3^4 + \frac{\gamma_3}{2} \theta_3^2 + \frac{1}{2}, \\
G_j & = \begin{cases} g_1, & j = 1 \\ g_2 d_1, & j = 2 \\ g_3 d_2, & j = 3 \end{cases}.
\end{aligned} \quad (76)$$

Define $\lambda_1 = \min \left\{ \lambda_{1,1}, 4 \left(k_2 - \frac{\bar{r}_1^2 g_{1M}^2}{4\sigma_e} \right), \gamma_{\theta_2}, \gamma_{b_2}, \lambda_{3,1} \right\}$ and $\lambda_2 = \sum_{j=1}^3 \lambda_{j,2}$. One can observe that both λ_1 and λ_2 are bounded. Then, (75) can be rewritten as

$$\mathcal{L}V \leq -\lambda_1 V + \lambda_2 + \sum_{j=1}^3 [1 + G_j N_j(\bar{\beta}_j)] \dot{\bar{\beta}}_j. \quad (77)$$

By applying Lemma 4, one can conclude that

$$\mathbb{E}[V(\mathcal{W})] \leq \mathbb{E}[V(\mathcal{W}(0))] e^{-\lambda_1 t} + \bar{\mathcal{E}} + \frac{\lambda_2}{\lambda_1} \quad (78)$$

with

$$\left| \int_0^t e^{-\lambda_1(t-\tau)} \sum_{j=1}^3 [G_j N_j(\bar{\beta}_j) + 1] \dot{\bar{\beta}}_j d\tau \right| \leq \bar{\mathcal{E}} \quad (79)$$

which implies that $\mathbb{E}(V)$ is bounded.

Denote that $V(\mathcal{W}) = V_c(\eta_1) + \bar{V}(\bar{\mathcal{W}})$ with

$$\bar{V}(\bar{\mathcal{W}}) = V_d(z_1, \bar{\theta}_1, \bar{a}_2) + V_2(e_2, \bar{\theta}_2, \bar{b}_2) + V_3(e_3, \bar{\theta}_3),$$

where $\bar{\mathcal{W}} = [e_2, e_3, z_1, \bar{\theta}_1, \bar{\theta}_2, \bar{\theta}_3, \bar{a}_2, \bar{b}_2]^\top$. With the fact that V_c and $\bar{V}$ are both positive definite, from (78), one can further conclude that $\mathbb{E}[V_c(\eta_1)]$ and $\mathbb{E}[\bar{V}(\bar{\mathcal{W}})]$ are also bounded for all $t \in [0, t_f]$ such that

$$\begin{aligned}
\mathbb{E}[V_c(\eta_1)] & \leq V(\mathcal{W}(0)) e^{-\lambda_1 t} + \bar{\mathcal{E}} + \frac{\lambda_2}{\lambda_1}, \\
\mathbb{E}[\bar{V}(\bar{\mathcal{W}})] & \leq V(\mathcal{W}(0)) e^{-\lambda_1 t} + \bar{\mathcal{E}} + \frac{\lambda_2}{\lambda_1}.
\end{aligned} \quad (80)$$

Furthermore, based on the definition of $V_c(\eta_1)$, one has $|e_1(t)| \neq k_b$ and $|e_1(t)| \neq k_a$. Then, the above inequality becomes

$$\mathbb{E}(\eta_1^4) \leq 1 - e^{-4V(\mathcal{W}(0))e^{-\lambda_1 t} - 4\bar{\mathcal{E}} - 4\frac{\lambda_2}{\lambda_1}}, \quad (81)$$

which implies that

$$\begin{aligned}
e_1(t) \in \bar{\Omega}_{e_1} & = \left\{ e_1 \left| -k_a \sqrt{1 - e^{-4V(\mathcal{W}(0))e^{-\lambda_1 t} - \frac{4\lambda_2}{\lambda_1} - 4\bar{\mathcal{E}}}} \right. \right. \\
& \leq \mathbb{E}(e_1) \leq k_b \sqrt{1 - e^{-4V(\mathcal{W}(0))e^{-\lambda_1 t} - \frac{4\lambda_2}{\lambda_1} - 4\bar{\mathcal{E}}}} \left. \right\}.
\end{aligned}$$

Hence, given the original value $e_1(0)$ satisfying $-k_a(0) < \eta_1(0) < k_b(0)$, one can conclude that $\mathbb{E}[e_1(t)]$ remains in the set $\bar{\Omega}_{e_1} \subset (-k_a(t), k_b(t))$ for all $t \in [0, t_f]$.

Accordingly, $\bar{\mathcal{W}}$ is bounded due to (80). Note that $\hat{\theta}_i = \theta_i - \bar{\theta}_i$, $\hat{a}_2 = a_2 - \bar{a}_2$, and $\hat{b}_2 = b_2 - \bar{b}_2$, one can conclude that the adaptive compensation laws designed in Section 4 are bounded. Since $x_1 = e_1 + x_d$, the boundedness of x_1 can be obtained as e_1 is bounded and the reference signal satisfies Assumption 2. $x_3 = e_3$ is bounded. Also, the DSC filter error z_1 and the virtual control signal s_1 in (30) are bounded with the boundedness of the adaptive Nussbaum gains β_j and the DSC internal state α_1 . Finally, the control signals v_q in (36) and v_d in (40) are also bounded. Therefore, all signals in the closed-loop are bounded and the asymmetric constraints (26) of $e_1(t)$ are never violated. This completes the proof. $\square$

Table 1
PMSM parameters.

R (Ω)	L_d (H)	L_q (H)	ψ_f (Web)	J (kg m ²)	P_n	B
1.55	0.014	0.052	0.596	0.2	2	0.0005

Table 2
PMSM state space model.

a_1	a_2	a_3	a_4	b_1	b_2	b_3
-0.003	-1.143	-10	17.886	-29.877	-0.265	-11.492
b_4	c_1	c_2	c_3	h_1	h_2	h_3
19.275	-112.56	3.768	72.622	10	19.275	72.622

6. Simulation studies

In this section, to verify the proposed scheme in Section 4, we implement the adaptive Nussbaum controller design on the stochastic PMSM system (20). Specifically, we consider the parameters of PMSM as listed in Table 1. Accordingly, the parameters in (22) can be calculated as listed in Table 2.

As shown in (21), the dead-zone mapping is considered separately for u_d and u_q as

$$u_q = \begin{cases} v_q - 3, & v_q \in [3, +\infty) \\ 0, & v_q \in (-3, 3) \\ v_q + 3, & v_q \in (-\infty, -3] \end{cases},$$

$$u_d = \begin{cases} \cos(v_d)(v_d - 4), & v_d \in [4, +\infty) \\ 0, & v_d \in (-4, 4) \\ \cos(v_d)(v_d + 4), & v_d \in (-\infty, -4] \end{cases}. \quad (82)$$

In addition, to testify the robustness of the adaptive Nussbaum design, the time-varying load is selected as follows

$$T_L = \begin{cases} 1, & t \in [3, 4) \\ 0, & \text{otherwise} \end{cases}. \quad (83)$$

In the tracking problem, the reference signal is considered as

$$x_d = 0.75 \sin(3t) + 0.25 \sin(4t). \quad (84)$$

For the time-varying constraints (26), the performance functions are selected as

$$k_a(t) = \frac{1}{t+1}, \quad k_b(t) = \frac{1}{t+2}. \quad (85)$$

The RBFNNs are selected as Gaussian functions in the form of (18). Specifically, the RBFNN $\varphi_1(x_1, r_1, \eta_1)$, $\varphi_2(x_1, x_2, e_2)$ and $\varphi_3(x_1, x_1, e_3)$ in (49), (59) and (68) are all composed of 27 nodes with the center spaced evenly in the region $[-2, 0, 2] \times [-2, 0, 2] \times [-2, 0, 2]$ and the width being selected as 1. To determine the control parameters, according to (80), the term $e^{-\lambda_1 t}$ directly affects the convergence speed of V_c and $\bar{V}$, while the term $\frac{\lambda_2}{\lambda_1}$ influences the upper bound of the convergence interval. Therefore, to increase the convergence speed and decrease the convergence region of V_c and $\bar{V}$, one should enlarge the value of λ_1 and decrease the value of λ_2 .

For the first subsystem, we investigate the virtual control and adaptive compensation laws as (30) and (32), respectively in which the parameters are designed as $k_1 = 200$, $\gamma_{\theta_1} = 5$, $\gamma_{a_1} = 5$, and the time constant $\tau_1 = 3$ of DSC filter. The simulation results are shown in Figs. 2 and 3. The tracking process is shown in Fig. 2-a, where the electric angular velocity $x_1 = \omega_e$ can track the reference signal x_d even there exists load changes. The time-varying constraints on the output tracking error e_1 is shown in Fig. 2-b, which are never violated during the online control. As discussed in Section 4, the time-varying asymmetric constraints on the error e_1 can be transformed as the time-invariant symmetric constraints on the error η_1 . From Fig. 2-c, the transformed error η_1 always stays in the range of $(-1, 1)$, which conforms with the

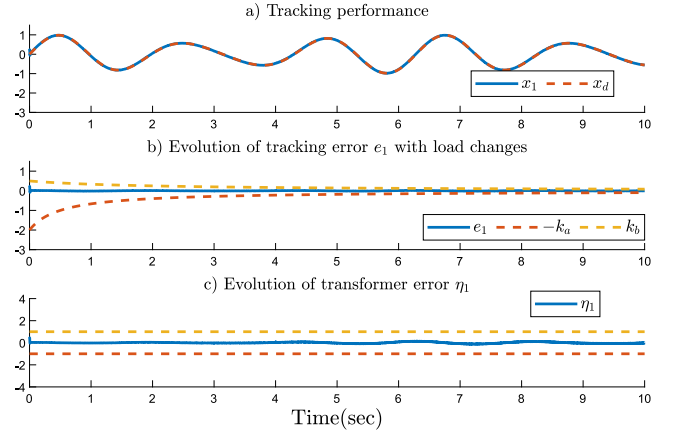


Fig. 2. Evolutions of the state x_1 and the errors e_1 , η_1 in the first subsystem.

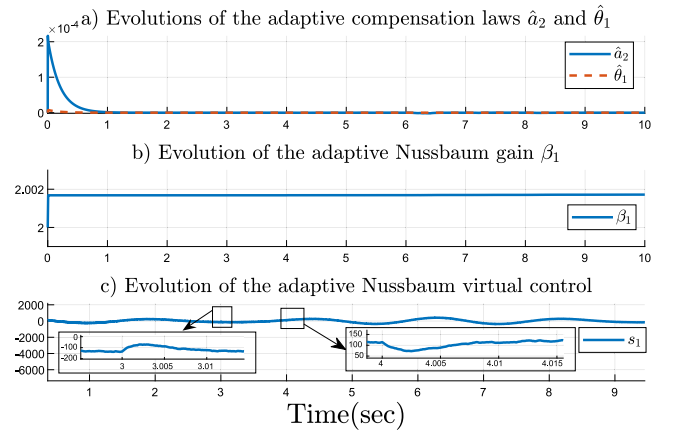


Fig. 3. Evolutions of the adaptive signals in the first subsystem.

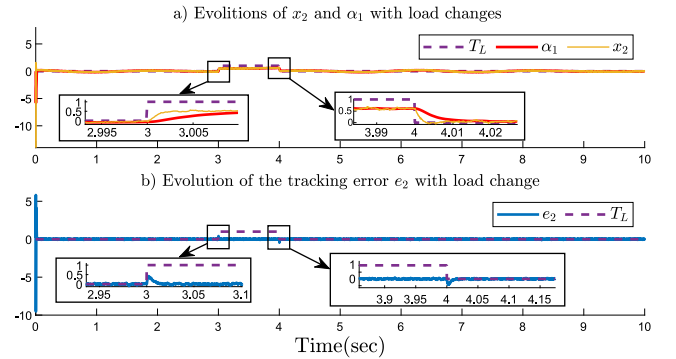


Fig. 4. Evolutions of the state x_2 and the tracking error e_2 in the second subsystem.

fact that e_1 is within the range $(-k_a, k_b)$ as in Fig. 2-b. The adaptive Nussbaum design is shown in Fig. 3, from which one can observe that the Nussbaum-based design can counteract the effect of load changes. In addition, all adaptive signals are shown to converge to bounded values within one second.

For the second subsystem, we propose the control input v_q in (36) and the adaptive compensation laws in (38). In terms of concreteness, the parameters are $k_2 = 200$, $\gamma_{\theta_2} = 5$, and $\gamma_{b_2} = 5$. Based on these settings, the simulation results are shown in Figs. 4 and 5. From Fig. 4, one can observe that the q -axis current $x_2 = i_q$ can track the internal input signal α_1 of DSC filter. Hence, the tracking error e_2 converges to the neighborhood of origin despite of the existence of time-varying

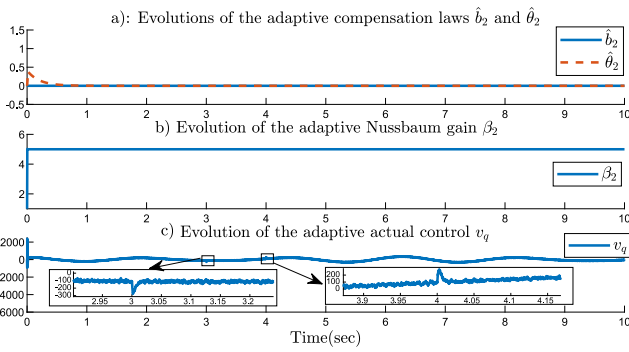


Fig. 5. Evolutions of the adaptive signals in the second subsystem.

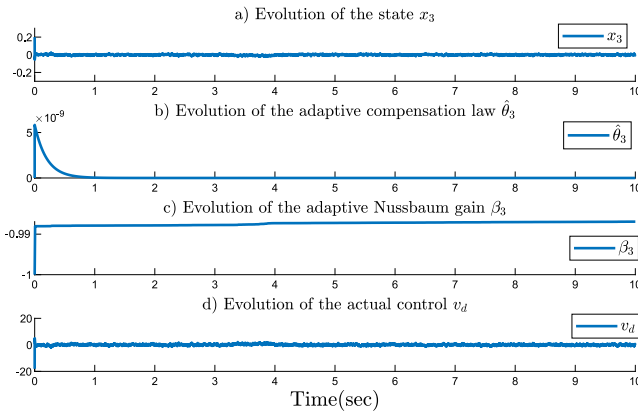


Fig. 6. Evolutions of the signals in the third subsystem.

load and stochastic disturbance. Fig. 5 indicates the effectiveness of the proposed adaptive Nussbaum design.

For the third subsystem, consider the control input v_d designed in (40) and adaptive compensation law in (42) with the parameters $k_3 = 200$ and $\gamma_{\theta_3} = 5$. Finally, in Fig. 6, one can observe that the d -axis current $x_3 = i_d$ converges toward origin. Moreover, the adaptive signals $\hat{\theta}_3$ and β_3 remain bounded.

7. Conclusions

For PMSM, it is challenging to take the time-varying output constraints with input dead-zone subject to multiple unknown control gains under stochastic disturbance into account simultaneously. To tackle this issue, a novel adaptive Nussbaum backstepping design is proposed. The Nussbaum-based Lyapunov fundamental lemma is established to decouple the multiple unknown control gains of the PMSM system. Applying the BLF with adaptive compensation laws and control signals, the PMSM angular velocity can track the desired trajectory and the tracking error never violates the given asymmetric time-varying constraints with all signals in the closed-loop system being bounded. Simulation studies have been conducted to verify the effectiveness of the presented control scheme. To enhance the efficiency of the proposed method while reducing computational complexity and communication load in real-time applications, future work will focus on extending the current research to event-triggered feedback design.

CRediT authorship contribution statement

Tianhao Fei: Writing – original draft, Software, Methodology, Investigation. **Yongliang Yang:** Writing – original draft, Supervision, Conceptualization. **Chunyu Zheng:** Validation, Investigation. **Hu Zhang:** Validation. **Guofeng Yuan:** Validation, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] Y. Wang, H. Yu, Y. Liu, Speed-current single-loop control with overcurrent protection for PMSM based on time-varying nonlinear disturbance observer, *IEEE Trans. Ind. Electron.* 69 (1) (2022) 179–189.
- [2] S. Zhou, Y. Li, S. Tong, Finite-time adaptive neural network event-triggered output feedback control for PMSMs, *Neurocomputing* 533 (7) (2023) 10–21.
- [3] S. Aouaouda, M. Chadli, M. Boukhni, Speed sensor fault tolerant controller design for induction motor drive in EV, *Neurocomputing* 214 (19) (2016) 32–43.
- [4] R. Jon, Z. Wang, C. Luo, M. Jong, Adaptive robust speed control based on recurrent elman neural network for sensorless PMSM servo drives, *Neurocomputing* 227 (1) (2017) 131–141.
- [5] P. Mani, R. Rajan, L. Shanmugam, Y.H. Joo, Adaptive fractional fuzzy integral sliding mode control for PMSM model, *IEEE Trans. Fuzzy Syst.* 27 (8) (2018) 1674–1686.
- [6] L. Ding, Y.W. Li, N.R. Zargari, R. Paes, Sensorless control of CSC-fed PMSM drives with low switching frequency for electrical submersible pump application, *IEEE Trans. Ind. Appl.* 56 (4) (2020) 3799–3808.
- [7] Y. Chen, B. Zang, H. Wang, H. Liu, H. Li, Composite PM rotor design and alternating flux density harmonic component analysis of a 200 kW high-speed PMSM used in FESS, *IEEE Trans. Ind. Appl.* 59 (2) (2023) 1469–1480.
- [8] S. Cheng, J. Yu, L. Zhao, Y. Ma, Adaptive fuzzy control for permanent magnet synchronous motors considering input saturation in electric vehicle stochastic drive systems, *J. Franklin Inst.* 357 (13) (2020) 8473–8490.
- [9] G. Li, X. Chen, J. Yu, J. Liu, Adaptive neural network-based finite-time impedance control of constrained robotic manipulators with disturbance observer, *IEEE Trans. Circuits Syst. II: Express Briefs* 69 (3) (2022) 1412–1416.
- [10] Y. Shi, W. Xie, G. Zhang, W. Zhang, C. Silvestre, Event-triggered quantitative prescribed performance neural adaptive control for autonomous underwater vehicles, *IEEE Trans. Syst. Man Cybern.: Syst.* 54 (6) (2024) 3381–3392.
- [11] Y. Song, G. Cheng, Y. Tuo, Adaptive output feedback control for chaotic PMSMs stochastic system considering constraints, *Chaos Solitons Fractals* 187 (2024) 115321.
- [12] J. Wu, W. Xu, X. Wang, R. ru Ma, Stochastic adaptive fixed-time stabilization of chaotic systems with applications in PMSM and FWS, *Chaos Solitons Fractals* 153 (2021) 111582.
- [13] S. Cheng, J. Yu, C. Lin, L. Zhao, Y. Ma, Neuroadaptive finite-time output feedback control for PMSM stochastic nonlinear systems with iron losses via dynamic surface technique, *Neurocomputing* 402 (18) (2020) 162–170.
- [14] H.-B. Ji, H.-S. Xi, Adaptive output-feedback tracking of stochastic nonlinear systems, *IEEE Trans. Autom. Control* 51 (2) (2006) 355–360.
- [15] H. Wang, B. Chen, K. Liu, X. Liu, C. Lin, Adaptive neural tracking control for a class of nonstrict-feedback stochastic nonlinear systems with unknown backlash-like hysteresis, *IEEE Trans. Neural Netw. Learn. Syst.* 25 (5) (2014) 947–958.
- [16] X. Ye, J. Jiang, Adaptive nonlinear design without a priori knowledge of control directions, *IEEE Trans. Autom. Control* 43 (11) (1998) 1617–1621.
- [17] Y. Yang, G. Liu, Q. Li, C.K. Ahn, Multiple adaptive fuzzy Nussbaum-type functions design for stochastic nonlinear systems with fixed-time performance, *Fuzzy Sets and Systems* 476 (30) (2024) 108767.
- [18] C. Chen, Z. Liu, Y. Zhang, C.L.P. Chen, S. Xie, Saturated Nussbaum function based approach for robotic systems with unknown actuator dynamics, *IEEE Trans. Cybern.* 46 (10) (2016) 2311–2322.
- [19] W. Chen, X. Li, W. Ren, C. Wen, Adaptive consensus of multi-agent systems with unknown identical control directions based on a novel Nussbaum-type function, *IEEE Trans. Autom. Control* 59 (7) (2014) 1887–1892.
- [20] C. Chen, Z. Liu, K. Xie, Y. Zhang, C.L.P. Chen, Asymptotic fuzzy neural network control for pure-feedback stochastic systems based on a semi-Nussbaum function technique, *IEEE Trans. Cybern.* 47 (9) (2017) 2448–2459.
- [21] H. Ma, H. Ren, Q. Zhou, R. Lu, H. Li, Approximation-based Nussbaum gain adaptive control of nonlinear systems with periodic disturbances, *IEEE Trans. Syst. Man Cybern.: Syst.* 52 (4) (2022) 2591–2600.
- [22] C. Wen, J. Zhou, Z. Liu, H. Su, Robust adaptive control of uncertain nonlinear systems in the presence of input saturation and external disturbance, *IEEE Trans. Autom. Control* 56 (7) (2011) 1672–1678.
- [23] J. Wang, H. Zhang, K. Ma, Z. Liu, C.L.P. Chen, Neural adaptive self-triggered control for uncertain nonlinear systems with input hysteresis, *IEEE Trans. Neural Netw. Learn. Syst.* 33 (11) (2022) 6206–6214.

- [24] B. Ren, S.S. Ge, K.P. Tee, T.H. Lee, Adaptive neural control for output feedback nonlinear systems using a barrier Lyapunov function, *IEEE Trans. Neural Netw.* 21 (8) (2010) 1339–1345.
- [25] L. Fang, L. Ma, S. Ding, J.H. Park, Finite-time stabilization of high-order stochastic nonlinear systems with asymmetric output constraints, *IEEE Trans. Syst. Man Cybern.: Syst.* 51 (11) (2021) 7201–7213.
- [26] J. Zhang, S. Wang, P. Zhou, L. Zhao, S. Li, Novel prescribed performance-tangent barrier Lyapunov function for neural adaptive control of the chaotic PMSM system by backstepping, *Int. J. Electr. Power Energy Syst.* 121 (2020) 105991.
- [27] L. Gao, Z. Song, Z. Wang, P. Li, Finite-time adaptive neural control of uncertain constrained nonlinear systems with actuator fault, *Mech. Syst. Signal Process.* 200 (1) (2023) 110555.
- [28] A. Boulkroune, A fuzzy adaptive control approach for nonlinear systems with unknown control gain sign, *Neurocomputing* 179 (29) (2016) 318–325.
- [29] H. Wang, X. Liu, K. Liu, Adaptive neural databased compensation control of nonlinear systems with dynamic uncertainties and input saturation, *IET Control Theory Appl.* 9 (7) (2015) 1058–1065.
- [30] X. Wang, X. Wang, Z. Wang, X. Xiao, S. Li, Composite finite-time control for PMSM with prescribed performance using disturbance compensation technique, *Control Eng. Pract.* 141 (2023) 105677.
- [31] Y.-F. Gao, X.-M. Sun, C. Wen, W. Wang, Adaptive tracking control for a class of stochastic uncertain nonlinear systems with input saturation, *IEEE Trans. Autom. Control* 62 (5) (2017) 2498–2504.
- [32] Y.-J. Liu, S. Lu, S. Tong, X. Chen, C.P. Chen, D.-J. Li, Adaptive control-based barrier Lyapunov functions for a class of stochastic nonlinear systems with full state constraints, *Automatica* 87 (2018) 83–93.
- [33] C. Qian, W. Lin, Non-Lipschitz continuous stabilizers for nonlinear systems with uncontrollable unstable linearization, *Systems Control Lett.* 42 (3) (2001) 185–200.
- [34] W. Si, X. Dong, Barrier Lyapunov function-based decentralized adaptive neural control for uncertain high-order stochastic nonlinear interconnected systems with output constraints, *J. Franklin Inst.* 355 (17) (2018) 8484–8509.
- [35] Y. Wei, H. Wang, Y. Tian, Adaptive time-varying barrier Lyapunov function-based model-free hybrid position/force control for series elastic actuator-based manipulator, *IEEE Trans. Circuits Syst. II: Express Briefs* 71 (1) (2024) 271–275.
- [36] Y.-H. Liu, Y. Liu, Y.-F. Liu, C.-Y. Su, Adaptive fuzzy control with global stability guarantees for unknown strict-feedback systems using novel integral barrier Lyapunov functions, *IEEE Trans. Syst. Man Cybern.: Syst.* 52 (7) (2022) 4336–4348.
- [37] Z. Yu, S. Li, Neural-network-based output-feedback adaptive dynamic surface control for a class of stochastic nonlinear time-delay systems with unknown control directions, *Neurocomputing* 129 (10) (2014) 540–547.
- [38] K.P. Tee, S.S. Ge, E.H. Tay, Barrier Lyapunov Functions for the control of output-constrained nonlinear systems, *Automatica* 45 (4) (2009) 918–927.
- [39] G. Liu, Y. Yang, X. Zhao, C.K. Ahn, Adaptive fuzzy practical bipartite synchronization for multiagent systems with intermittent feedback under multiple unknown control directions, *IEEE Trans. Fuzzy Syst.* 32 (5) (2024) 3211–3225.
- [40] D. Wang, J. Huang, Neural network-based adaptive dynamic surface control for a class of uncertain nonlinear systems in strict-feedback form, *IEEE Trans. Neural Netw.* 16 (1) (2005) 195–202.
- [41] J. Na, Adaptive prescribed performance control of nonlinear systems with unknown dead zone, *Internat. J. Adapt. Control Signal Process.* 27 (5) (2013) 426–446.
- [42] S. Ge, F. Hong, T.H. Lee, Adaptive neural control of nonlinear time-delay systems with unknown virtual control coefficients, *IEEE Trans. Syst. Man Cybern. B* 34 (1) (2004) 499–516.
- [43] Y. Yang, L. Tang, W. Zou, C.K. Ahn, Novel command-filtered Nussbaum design for continuous-time nonlinear dynamical systems with multiple unknown high-frequency gains, *Nonlinear Dynam.* 111 (2023) 4313–4323.



Tianhao Fei received his B.E. degree from University of Science and Technology Beijing, Beijing, China, in 2024. He is currently pursuing the M.E. degree in Control Science and Technology at University of Science and Technology Beijing, Beijing. His current research interests include intelligent control for robots.



Yongliang Yang received the B.S. degree in electrical engineering from Hebei University, Baoding, China, in 2011, and the Ph.D. degree in control theory and control engineering from the University of Science and Technology Beijing (USTB), Beijing, China, in 2018.

From 2015 to 2017, he was a Visiting Scholar with the Missouri University of Science and Technology, Rolla, MO, USA, sponsored by China Scholarship Council. He was an Assistant Professor with USTB from 2018 to 2020. From 2020 to 2021, he was an independent Postdoctoral Research Fellow with the State Key Laboratory of Internet of Things for Smart City, Faculty of Science and Technology, University of Macau. He is currently an Associate Professor with USTB. His research interests include reinforcement learning theory, robotics, distributed optimization and control for cyberphysical systems.

Dr. Yang was a recipient of the Best Ph.D. Dissertation of the China Association of Artificial Intelligence, the Best Ph.D. Dissertation of USTB, Chancellor's Scholarship in USTB, Excellent Graduates Awards in Beijing, and UM Macao Talent Program in Macau. He is an Associate Editor for *IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS*.

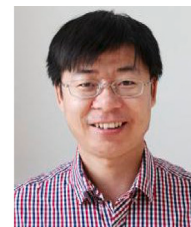


Chunyu Zheng was born in Fugou, Henan, China, in 1987. He received the B.S. degree from China University of Mining and Technology, Beijing, in 2008 and the M.S. degree from North China University of Technology (NCUT), Beijing, in 2011, both in electrical engineering.

He is currently working in NCUT. His research interests include power electronic converter, position/speed sensorless control of ac motors and high-performance traction drives, and electric vehicle.



Hu Zhang received his B.E. degree from Hebei University of Technology, Tianjin, China, in 2000, and the Ph.D. degree in control theory and control engineering from the University of Science and Technology Beijing, Beijing, China, in 2011. He is currently an Associate Professor with the Department of Electrical Engineering of North China University of Technology, Beijing. His research interests include the control of electrical machines and parameter identification.



Guofeng Yuan (Member, IEEE) was born in Heilongjiang China, in 1979. He received the B.S. and Ph.D. degrees in electrical engineering from the Tsinghua University, Beijing, China, in 2001 and 2006, respectively. From 2006 to 2008, he was an Assistant Professor with the Department of Electrical Engineering, Tsinghua University, Beijing. From 2008 to 2011, he was Chief Engineer with the Beijing Chino-Harvest Wind Power Technology Co. Ltd, Beijing. From 2011 to 2014, he was CTO with the Dalian GUOTONG Electrical Co. Ltd., Dalian, China. Now he is a professor with the Department of Electrical Engineering of North China University of Technology, Beijing. His research interests include the control of electrical machines and converters for wind power generation systems.